Worksheet 3: Introductory PostGIS Spatial Functions

Toy geometry dataset with points, linestrings and polygons

Goals. Practise very simple spatial queries in PostGIS: reading coordinates, computing length and area, testing containment and intersection, using envelopes / MBRs, comparing bounding-box interaction with exact predicates, and experimenting with DE-9IM using ST_Relate.

1. Dataset setup

Load the attached SQL script first. All coordinates are abstract planar units, so measurements are in the same arbitrary unit used in the picture.

```
\i simple_spatial_dataset_v2.sql

SELECT name, ST_AsText(geom) FROM demo_point ORDER BY name;
SELECT name, ST_AsText(geom) FROM demo_line ORDER BY name;
SELECT name, ST_AsText(geom) FROM demo_polygon ORDER BY name;
```

2. Illustration of the dataset

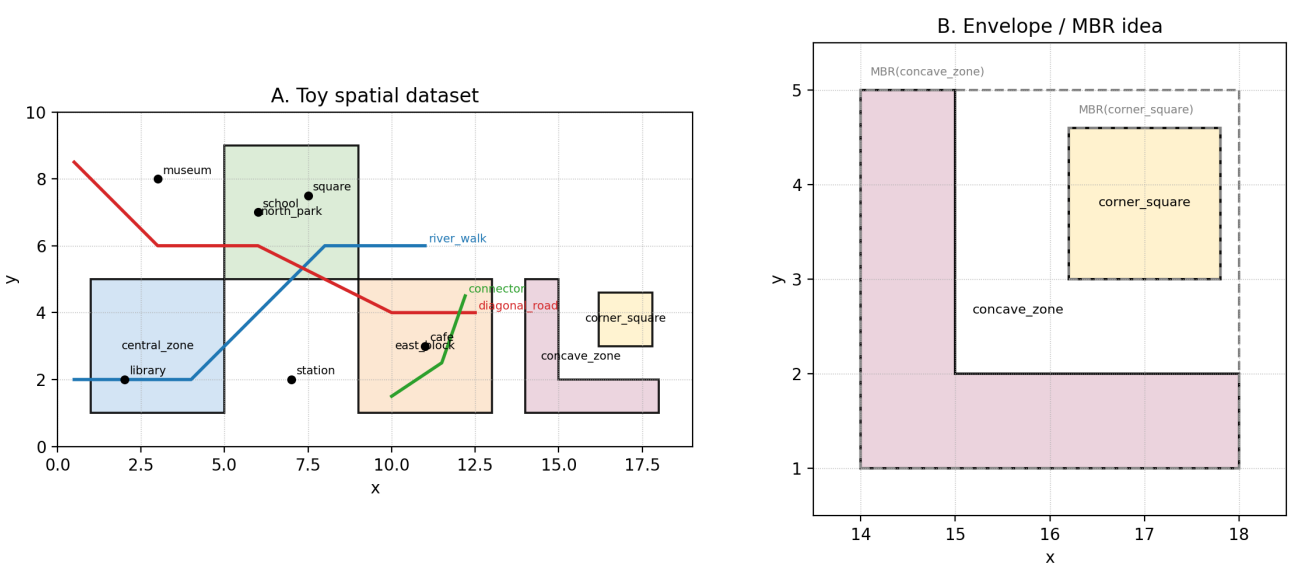

Figure 1. Main dataset (left) and a zoom used for the envelope / MBR exercises (right).

Reminder. The minimum bounding rectangle (MBR) of a geometry is the smallest axis-aligned rectangle that contains it. In PostGIS, ST_Envelope(geom) returns that rectangle as a geometry. The operator && compares bounding boxes, not the full exact geometries.

3. Official documentation to consult

Students are encouraged to use the official PostGIS online manual while solving the exercises.

- General function reference: https://postgis.net/docs/reference.html
- Usage chapter (relationships, DE-9IM, indexes and bounding-box operators): https://postgis.net/docs/using_postgis_dbmanagement.html

Useful functions for this worksheet include: ST_AsText, ST_GeometryType, ST_X, ST_Y, ST_Length, ST_Area, ST_Perimeter, ST_Centroid, ST_Contains, ST_Within, ST_Touches, ST_Disjoint, ST_Intersects, ST_Distance, ST_ConvexHull, ST_Envelope, ST_Relate and ST_RelateMatch.

4. Exercises

Exercise 1. Inspect the dataset

List all objects from the three tables and show their WKT representation with ST_AsText. For each row, also show the geometry type returned by ST_GeometryType.

Exercise 2. Read point coordinates

Return each point name together with ST_X(geom) and ST_Y(geom). Order the result by name.

Exercise 3. Measure the lines

For every row in demo_line, compute the line length with ST_Length. Which line is the longest?

Exercise 4. Measure the polygons

For every row in demo_polygon, compute area with ST_Area and perimeter with ST_Perimeter. Order the result by area descending.

Exercise 5. Count vertices

Use ST_NPoints (or ST_NumPoints when appropriate) to count the vertices of every line and every polygon.

Exercise 6. Points inside polygons

List every pair (point, polygon) such that the point is inside the polygon. Use an explicit spatial predicate.

Exercise 7. Compare contains and within

Write two equivalent queries: one using ST_Contains(polygon.geom, point.geom) and another using ST_Within(point.geom, polygon.geom). Confirm that they return the same result for this dataset.

Exercise 8. Intersections involving lines

List every pair (line, polygon) such that the line intersects the polygon. Then identify which pairs only touch a boundary and which pairs truly cross the polygon interior.

Exercise 9. Touches and disjointness between polygons

Using only rows from demo_polygon, find one pair of polygons that touch and one pair that are disjoint. Return the names and the Boolean result.

Exercise 10. Distances

Compute the distance from every point to east_block. Then compute the distance from station to each line. Identify the minimum distance in each query.

Exercise 11. Centroids

Return the centroid of every polygon with ST_Centroid, showing the result in WKT.

Exercise 12. Insert a new point

Insert a point named playground at coordinates (10.5, 3.5). Then determine in which polygon it lies and its distance to connector.

Exercise 13. Convex hull of all points

Use ST_Collect over demo_point and apply ST_ConvexHull. Return the hull in WKT. Bonus: compute its area.

Exercise 14. Envelope / MBR of each geometry

For every polygon, return ST_AsText(ST_Envelope(geom)). Explain, in words, what this envelope represents.

Exercise 15. Concave polygon versus its envelope

For concave_zone, compare ST_Area(geom) with ST_Area(ST_Envelope(geom)). How much empty space inside the MBR is not part of the actual polygon?

Exercise 16. MBR interaction using &&

Test the pair (concave_zone, corner_square) with both the operator && and the function ST_Intersects. Explain why the answers are different.

Exercise 17. Which pairs are filtered by MBR first?

List every pair of polygons whose MBRs intersect using the operator &&. Then run the same search with ST_Intersects. Which pairs survive the exact test?

Exercise 18. DE-9IM matrix for a touching case

Use ST_Relate to obtain the 9-character matrix for (central_zone, north_park). Confirm separately with ST_Touches whether the named predicate agrees with your interpretation.

Exercise 19. DE-9IM matrix for a disjoint case

Use ST_Relate to obtain the 9-character matrix for (concave_zone, corner_square). Confirm separately with ST_Disjoint.

Exercise 20. Pattern matching with ST_RelateMatch

Take one matrix produced by ST_Relate and use ST_RelateMatch to test a pattern of your choice from the manual. State the pattern you used and what relationship it captures.

Exercise 21. Aggregate exercise

For each polygon, count how many points it contains. Return polygon name and number of contained points, including polygons that contain zero points.

Exercise 22. Insert a new polygon

Insert a polygon named south_strip with WKT POLYGON((4 0, 10 0, 10 1, 4 1, 4 0)). Then test which existing polygons it intersects or touches.

Exercise 23. Envelope after insertion

After inserting south_strip, compute its envelope and compare it with the polygon itself. Does the envelope add any empty space in this case? Why or why not?

Exercise 24. Optional indexing question

Create a GiST index on each geometry column if it is not already present. Then use EXPLAIN on a query with ST_Intersects and another with &&. What do you observe?

5. Short hints

- When a query compares two tables spatially, remember that you can use a JOIN between them.
- ST_Envelope returns a geometry; use ST_AsText if you want to display it as text.
- The operator && compares bounding boxes. ST_Intersects compares the actual geometries.
- ST_Relate returns a 9-character DE-9IM matrix. ST_RelateMatch checks whether such a matrix fits a given pattern.